

One tensor to bracket them all

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- In geometric mechanics, **Poisson and Jacobi brackets** arise from different structures (symplectic, contact, cosymplectic...). Our goal is to **unify all these brackets** under a single framework.
- We consider a **non-degenerate 2-covariant tensor field** B with skew-symmetric part ω and symmetric part S .
- If $TM = \mathcal{H} \oplus \mathcal{V}$, with $\mathcal{H} = \text{Ker } S$ and $\mathcal{V} = \text{Ker } \omega$, we can give a formula which gives the exact obstruction for B to define a Poisson or Jacobi bracket:

$$\frac{1}{2} \llbracket \omega, \omega \rrbracket = -(\text{d}\omega)|_{\mathcal{H}} - \iota_{\text{curv}_{\mathcal{H}}} S.$$

2-covariant tensors

Consider a finite-dimensional vector space E and a 2-covariant tensor

$$B \in E^* \otimes E^* .$$

This tensor is **not necessarily symmetric nor skew-symmetric**, so we have two (different, in general) naturally induced linear maps:

$$\widehat{B}: E \longrightarrow E^* \quad \text{defined by} \quad \widehat{B}(v) = B(v, \cdot)$$

and

$${}^t\widehat{B}: E \longrightarrow E^* \quad \text{defined by} \quad {}^t\widehat{B}(v) = B(\cdot, v)$$

2-covariant tensors

We will denote by tB the unique 2-covariant tensor such that

$$\widehat{{}^tB} = \widehat{B}$$

Proposition

Any 2-covariant tensor $B \in E^ \otimes E^*$ can be decomposed canonically and uniquely into its skew-symmetric and symmetric parts*

$$B = \omega + S,$$

where $\omega \in \bigwedge^2 E^$ and $S \in S^2(E^*)$.*

Explicitly,

$$\omega = \frac{B - {}^tB}{2}, \quad S = \frac{B + {}^tB}{2}.$$

Non-degenerate 2-covariant tensors

Definition

We say that a 2-covariant tensor B is **non-degenerate** if the induced linear map $\widehat{B}: E \rightarrow E^*$ (or, equivalently, its transpose) is an isomorphism.

With this, there is a unique 2-contravariant tensor, its **inverse** B^{-1} , s.t.

$$\widehat{B^{-1}} = \left(\widehat{B}\right)^{-1}$$

Symplectic vector spaces and inner products are two extremal cases of non-degenerate 2-covariant tensors, corresponding to B being purely skew-symmetric and purely symmetric (and positive-definite), respectively. Also, if g is an inner product and ω is any skew-symmetric 2-covariant tensor, the tensor $B = \omega + g$ is always non-degenerate.

Bivector of a 2-covariant tensor

Again, any 2-contravariant tensor decomposes in its skew-symmetric and symmetric parts uniquely $B^{-1} = \Lambda + \Sigma$:

- We have $\widehat{\Lambda} = -{}^t\widehat{B}^{-1} \circ \widehat{\omega} \circ \widehat{B}^{-1}$, and so for any $\alpha_1, \alpha_2 \in E^*$

$$\Lambda(\alpha_1, \alpha_2) = -\omega(\widehat{B}^{-1}(\alpha_1), \widehat{B}^{-1}(\alpha_2)).$$

- Similarly, $\widehat{\Sigma} = {}^t\widehat{B}^{-1} \circ \widehat{S} \circ \widehat{B}^{-1}$, and so for any $\alpha_1, \alpha_2 \in E^*$ we have

$$\Sigma(\alpha_1, \alpha_2) = S(\widehat{B}^{-1}(\alpha_1), \widehat{B}^{-1}(\alpha_2)).$$

If we denote by

$$\bigwedge^k \widehat{B}: \bigwedge^k E \longrightarrow \bigwedge^k E^*$$

the natural extensions of B to the exterior algebra then, we can write

$$\Lambda = -(\bigwedge^2 {}^tB^{-1})(\omega).$$

Asymmetry endomorphism

Definition

The **asymmetry** of a non-degenerate 2-covariant tensor B is the endomorphism:

$$\hat{J} := {}^t\hat{B}^{-1} \circ \hat{B}: E \longrightarrow E.$$

Equivalently,

$$B(u, \hat{J}(v)) = B(v, u),$$

for all $u, v \in E$. Hence,

$$B(\hat{J}(u), \hat{J}(v)) = B(u, v).$$

Also,

$$\hat{\omega} = \frac{1}{2}(\hat{B} - {}^t\hat{B}) = \frac{1}{2} {}^t\hat{B}(\hat{J} - \text{Id}), \quad \text{and} \quad \hat{S} = \frac{1}{2}(\hat{B} + {}^t\hat{B}) = \frac{1}{2} {}^t\hat{B}(\hat{J} + \text{Id}).$$

Asymmetry endomorphism

Thus, $\hat{J} = \text{Id}$ if and only if B is symmetric, and $\hat{J} = -\text{Id}$ if and only if B is skew-symmetric. More specifically,

$$v \in \text{Ker } \omega \iff \hat{J}(v) = v, \quad \text{and} \quad v \in \text{Ker } S \iff \hat{J}(v) = -v.$$

In particular,

- $\text{Ker } \omega$ and $\text{Ker } S$ are always in direct sum when B is non-degenerate.
- $E = \text{Ker } \omega \oplus \text{Ker } S$ if, and only if, J is an involution, so $\hat{J}^2 = \text{Id}$.

Remark

If E is even-dimensional and (ω, g) defines a Kähler structure, the asymmetry of $B = g + \omega$ corresponds to the induced complex structure J .

Direct sum decomposition

We will be particularly interested in the case where:

- $B = \omega + S$ is non-degenerate
- $E = \mathcal{H} \oplus \mathcal{V}$ with $\mathcal{H} := \text{Ker } S$ and $\mathcal{V} := \text{Ker } \omega$

In this case, we have the projections $\text{pr}_{\mathcal{H}}: E \rightarrow \mathcal{H}$ and $\text{pr}_{\mathcal{V}}: E \rightarrow \mathcal{V}$ and

$$\widehat{B}^{-1} \circ \widehat{\omega} = \text{pr}_{\mathcal{H}}, \quad \widehat{B}^{-1} \circ \widehat{S} = \text{pr}_{\mathcal{V}}.$$

In particular, the map associated to the bivector Λ simplifies to:

$$\widehat{\Lambda} = \text{pr}_{\mathcal{H}} \circ \widehat{B}^{-1}.$$

When studying the bivector Λ we can always reduce ourselves to this case:

Proposition

For any non-degenerate B , there exists a non-degenerate B' such that:

- 1 $E = \text{Ker } \omega' \oplus \text{Ker } S'$ (Direct sum decomposition holds for B'),
- 2 $\Lambda = \Lambda'$ (skew-symmetric parts of B^{-1} and B'^{-1} coincide).

Brackets induced by a non-degenerate 2-covariant tensor

Now, we study 2-covariant tensor fields on a manifold M which satisfy the non-degeneracy condition point-wise. So,

$$B: M \rightarrow T^*M \otimes T^*M,$$

such that $B_x \in T_x^*M \otimes T_x^*M$ is a non-degenerate 2-covariant tensor.

In this case:

- The skew-symmetric part of B is a differential 2-form $\omega \in \Omega^2(M)$,
- The skew-symmetric part of B^{-1} is a bivector field $\Lambda \in \mathfrak{X}^2(M)$.

Aim of the work

When does the bivector field Λ define a Poisson or Jacobi structure?

Schouten–Nijenhuis bracket for Poisson and Jacobi

By results due to Lichnerowicz, to know if Λ defines a Poisson or Jacobi structure we need to compute $[\Lambda, \Lambda]$, where $[\cdot, \cdot]$ denotes the Schouten–Nijenhuis bracket.

Indeed, if $\Lambda \in \mathfrak{X}^2(M)$ is a bivector field:

- It is **Poisson** if and only if

$$[\Lambda, \Lambda] = 0.$$

And then the bracket is given by $\{f, g\} = \Lambda(df, dg)$.

- It is **Jacobi** if and only if there exists a v.f. $E \in \mathfrak{X}(M)$ such that

$$[\Lambda, \Lambda] = 2E \wedge \Lambda \text{ and } [\Lambda, E] = 0.$$

And then the bracket is given by $\{f, g\} = \Lambda(df, dg) + fE(g) - gE(f)$.

Conjugate Schouten–Nijenhuis bracket

To compute this, it will be useful to define the following bracket:

Definition (Conjugate Schouten–Nijenhuis bracket)

Let B be a non-degenerate 2-covariant tensor field. For any pair of forms $\alpha \in \Omega^a(M)$ and $\beta \in \Omega^b(M)$, the **conjugate Schouten–Nijenhuis bracket associated with B** , $\llbracket \cdot, \cdot \rrbracket : \Omega^a(M) \times \Omega^b(M) \longrightarrow \Omega^{a+b-1}(M)$, is given by

$$\llbracket \alpha, \beta \rrbracket := \wedge^{a+b-1} {}^t\widehat{B} \left(\left[\wedge^a {}^t\widehat{B}^{-1}(\alpha), \wedge^b {}^t\widehat{B}^{-1}(\beta) \right] \right).$$

With the conjugate bracket, the Poisson and Jacobi conditions on Λ , can be equivalently written, respectively, as

$$\llbracket \omega, \omega \rrbracket = 0 \quad \text{or} \quad \llbracket \omega, \omega \rrbracket = -2\alpha \wedge \omega \quad \text{and} \quad \llbracket \omega, \alpha \rrbracket = 0,$$

for some $\alpha \in \Omega^1(M)$, and then $E = {}^t\widehat{B}^{-1}(\alpha)$.

Formula for the conjugate Schouten–Nijenhuis bracket

Theorem

Let B be a non-degenerate 2-covariant tensor and let ω and S denote, respectively, its skew-symmetric and symmetric parts. Assume that $TM = \mathcal{H} \oplus \mathcal{V}$, with $\mathcal{H} := \text{Ker } S$ and $\mathcal{V} := \text{Ker } \omega$ vector subbundles. Then, for any $X, Y, Z \in \mathfrak{X}(M)$ we have

$$\begin{aligned} \frac{1}{2} \llbracket \omega, \omega \rrbracket(X, Y, Z) = & -d\omega(X_{\mathcal{H}}, Y_{\mathcal{H}}, Z_{\mathcal{H}}) - S([X_{\mathcal{H}}, Y_{\mathcal{H}}], Z_{\mathcal{V}}) \\ & - S([Y_{\mathcal{H}}, Z_{\mathcal{H}}], X_{\mathcal{V}}) - S([Z_{\mathcal{H}}, X_{\mathcal{H}}], Y_{\mathcal{V}}), \end{aligned}$$

where $X_{\mathcal{H}} = \text{pr}_{\mathcal{H}}(X)$ and $X_{\mathcal{V}} = \text{pr}_{\mathcal{V}}(X)$.

If the same conditions hold and $\beta \in \Omega^1(M)$, then

$$\llbracket \omega, \beta \rrbracket(X, Y) = (\mathcal{L}_{X_{\beta}} B)(X, Y_{\mathcal{H}}) - (\mathcal{L}_{X_{\beta}} B)(Y, X_{\mathcal{H}}) - (\mathcal{L}_{X_{\beta}} \omega)(X, Y),$$

where $X_{\beta} = {}^t \widehat{B}^{-1}(\beta)$.

Formula for the conjugate Schouten–Nijenhuis bracket

Let $TM = \mathcal{H} \oplus \mathcal{V}$ where $\mathcal{H} := \text{Ker } S$ and $\mathcal{V} := \text{Ker } \omega$.

The non-integrability of the distributions is measured by the vector-valued 2-forms $\text{curv}_{\mathcal{H}} \in \Omega^2(\mathcal{H}; \mathcal{V})$ and $\text{curv}_{\mathcal{V}} \in \Omega^2(\mathcal{V}; \mathcal{H})$:

$$\text{curv}_{\mathcal{H}}(X_{\mathcal{H}}, Y_{\mathcal{H}}) := \text{pr}_{\mathcal{V}}[X_{\mathcal{H}}, Y_{\mathcal{H}}], \quad \text{curv}_{\mathcal{V}}(X_{\mathcal{V}}, Y_{\mathcal{V}}) := \text{pr}_{\mathcal{H}}[X_{\mathcal{V}}, Y_{\mathcal{V}}].$$

Pairing $\text{curv}_{\mathcal{H}}$ with S defines a tensor $\iota_{\text{curv}_{\mathcal{H}}} S \in \Gamma(\wedge^2 \mathcal{H}^* \otimes \mathcal{V}^*)$:

$$(\iota_{\text{curv}_{\mathcal{H}}} S)(H_1, H_2, V_1) := S(\text{curv}_{\mathcal{H}}(H_1, H_2), V_1).$$

Theorem

Let $B = \omega + S$ be non-degenerate such that $TM = \mathcal{H} \oplus \mathcal{V}$. Then:

$$\frac{1}{2} \llbracket \omega, \omega \rrbracket = -(\text{d}\omega)|_{\mathcal{H}} - \iota_{\text{curv}_{\mathcal{H}}} S.$$

Local expression

As S is a symmetric tensor, it can be diagonalized locally. So

$$S = \sum_k \lambda_k \tau_k \otimes \tau_k,$$

for some local 1-forms τ_k , and $\lambda_k \in \mathbb{R}$.

We have $\mathcal{H} = \text{Ker } S = \bigcap \text{Ker } \tau_k$, and $\mathcal{V} = \text{Ker } \omega = \text{Span}(\Delta_1, \dots, \Delta_p)$, where $\{\Delta_k\}$ is the basis of $\mathcal{V}$ induced by the basis $\{\tau_k\}$ of $\mathcal{V}^*$, so

$$\text{curv}_{\mathcal{H}} = \sum_k \tau_k([H_1, H_2])\Delta_k = - \sum_k (d\tau_k)(H_1, H_2)\Delta_k,$$

for arbitrary H_1 and H_2 horizontal vector fields. Then, we have

$$\iota_{\text{curv}_{\mathcal{H}}} S = - \sum_k \lambda_k (d\tau_k)|_{\mathcal{H}} \wedge \tau_k,$$

and thus,

$$\frac{1}{2} \llbracket \omega, \omega \rrbracket = -(d\omega)|_{\mathcal{H}} + \sum_k \lambda_k (d\tau_k)|_{\mathcal{H}} \wedge \tau_k.$$

Now, we can use the theorem to characterize the conditions for a non-degenerate 2-covariant tensor field B to be Poisson or Jacobi.

Theorem (Poisson)

Assume that $TM = \mathcal{H} \oplus \mathcal{V}$, with $\mathcal{H} := \text{Ker } S$ and $\mathcal{V} := \text{Ker } \omega$. The bivector field Λ defines a Poisson structure iff the following conditions hold

- 1** $d\omega$ vanishes on horizontal vectors, so $(d\omega)|_{\mathcal{H}} = 0$.
- 2** *The horizontal distribution $\mathcal{H}$ is integrable.*

Theorem (Jacobi)

Assume that $TM = \mathcal{H} \oplus \mathcal{V}$, where $\mathcal{H} := \text{Ker } S$ and $\mathcal{V} := \text{Ker } \omega$ are vector subbundles. Then, $\Lambda = - \left(\bigwedge^2 {}^t B^{-1} \right) (\omega)$ can be completed to a Jacobi structure iff the following conditions hold

- 1 There exists a 1-form $\bar{\alpha} \in \mathcal{H}^*$ satisfying $(d\omega)|_{\mathcal{H}} = \bar{\alpha} \wedge (\omega|_{\mathcal{H}})$,
- 2 There exists a v.f. $R \in \Gamma(\mathcal{V})$ s.t. $\text{curv}_{\mathcal{H}}(X_{\mathcal{H}}, Y_{\mathcal{H}}) = -\omega(X_{\mathcal{H}}, Y_{\mathcal{H}})R$,
- 3 If $\alpha = \bar{\alpha} + {}^t \hat{B}(R)$ and $X_{\alpha} = {}^t \hat{B}^{-1}(\alpha)$, we have

$$(\mathcal{L}_{X_{\alpha}} \omega)|_{\mathcal{H}} = (\iota_{X_{\alpha}} d\omega - d\bar{\alpha})|_{\mathcal{H}} = 0 \quad \text{and} \quad [X_{\alpha}, \mathcal{H}] \subseteq \mathcal{H}.$$

The Jacobi structure is $\Lambda = - \left(\bigwedge^2 {}^t B^{-1} \right) (\omega)$ and $E = X_{\alpha} = {}^t \hat{B}^{-1}(\bar{\alpha}) + R$.

Symplectic structures

A **symplectic form** on a manifold M is a 2-form $\omega \in \Omega^2(M)$ such that $d\omega = 0$ and ω is non-degenerate. We can consider the 2-covariant tensor

$$B = \omega.$$

Since $\text{Ker } \omega = \{0\}$ and the symmetric part is $S = 0$, we have a trivial splitting $TM = \text{Ker } \omega \oplus \text{Ker } S = \text{Ker } S$, so $\mathcal{H} = TM$ and $\text{curv}_{\mathcal{H}} = 0$.

Applying the formula to compute the Schouten–Nijenhuis bracket:

$$\frac{1}{2}[\![\omega, \omega]\!] = -d\omega|_{\text{Ker } S} - \iota_{\text{curv}_{\mathcal{H}}} S = -d\omega = 0.$$

Therefore, the bivector field $\Lambda = -\left(\wedge^2 \iota_{\omega}^{-1}\right)(\omega)$ associated with ω defines a **Poisson structure**.

Locally conformally symplectic structures

A 2-form ω is **locally conformally symplectic (LCS)** if it is non-degenerate and there exists a closed 1-form $\theta \in \Omega^1(M)$ such that

$$d\omega = -\theta \wedge \omega.$$

If we let $B = \omega$, we have $\mathcal{H} = TM$ and $S = 0$. We have:

$$\frac{1}{2}[\omega, \omega] = -d\omega = \theta \wedge \omega.$$

Moreover, $\Delta = -\widehat{\omega}^{-1}(-\theta)$ is the Liouville vector field ($\iota_{\Delta}\omega = \theta$). So,

$$\frac{1}{2}[\omega, -\theta] = \mathcal{L}_{\Delta}\omega = \iota_{\Delta}d\omega + d(\iota_{\Delta}\omega) = -\iota_{\Delta}(\theta \wedge \omega) + d\theta = 0.$$

Thus, every LCS structure defines a **Jacobi structure** with $\Lambda = -\left(\wedge^2 \iota_{\omega}^{-1}\right)(\omega)$ and vector field $E = \widehat{\omega}^{-1}(-\theta) = \Delta$.

Contact structures

A **contact form** on M^{2n+1} is a 1-form η such that $\eta \wedge (d\eta)^n \neq 0$. This is equivalent to the 2-covariant tensor field

$$B = d\eta + \eta \otimes \eta,$$

being non-degenerate.

The distributions $\mathcal{H} = \text{Ker } \eta$ and $\mathcal{V} = \text{Ker } d\eta$ satisfy $TM = \mathcal{H} \oplus \mathcal{V}$, so:

$$\frac{1}{2} \llbracket d\eta, d\eta \rrbracket = -(d(d\eta))|_{\mathcal{H}} + (d\eta)|_{\mathcal{H}} \wedge \eta = d\eta \wedge \eta = -(-\eta) \wedge d\eta.$$

Since ${}^t\widehat{B}^{-1}(-\eta) = -R$, and $\mathcal{L}_R d\eta = \mathcal{L}_R \eta = 0$, we have

$$\frac{1}{2} \llbracket d\eta, -\eta \rrbracket = 0.$$

So every contact structure defines a **Jacobi structure** with

$$\Lambda = -\left(\wedge^2 {}^tB^{-1}\right)(d\eta) \text{ and } E = -R.$$

Almost cosymplectic structures of order p

An **almost cosymplectic structure of order p** on M^{2n+p} is a tuple $(\omega, \tau_1, \dots, \tau_p)$ such that

$$\omega^n \wedge \tau_1 \wedge \dots \wedge \tau_p \neq 0.$$

We can consider the non-degenerate 2-covariant tensor

$$B = \omega + \sum_{k=1}^p \tau_k \otimes \tau_k,$$

which satisfies $TM = \mathcal{H} \oplus \mathcal{V}$, with $\mathcal{H} = \bigcap \text{Ker } \tau_k$ and $\mathcal{V} = \text{Ker } \omega$.

We have that $\mathcal{V} = \text{Ker } \omega = \text{Span}(R_1, \dots, R_p)$, where R_k are the unique vector fields satisfying $\iota_{R_k} \omega = 0$ and $\tau_j(R_k) = \delta_{jk}$.

Almost cosymplectic structures of order p

For an almost cosymplectic structure of order p , we have:

- It defines a **Poisson structure** if and only if

$$(d\omega)|_{\mathcal{H}} = 0 \quad \text{and} \quad (d\tau_k)|_{\mathcal{H}} = 0, \quad \text{for every } k = 1, \dots, p.$$

- It defines a **Jacobi structure** if and only if there exist $\bar{\alpha} \in \Gamma(\mathcal{H}^*)$ and $\lambda_1, \dots, \lambda_p \in \mathcal{C}^\infty(M)$ satisfying:

1 $(d\omega)|_{\mathcal{H}} = \bar{\alpha} \wedge (\omega|_{\mathcal{H}})$ and $(d\tau_k)|_{\mathcal{H}} = \lambda_k(\omega|_{\mathcal{H}})$ for all k .

2 Let $X_\alpha = {}^t\widehat{B}^{-1}(\bar{\alpha}) - \sum_{k=1}^p \lambda_k R_k$. For every $k = 1, \dots, p$:

$$(\iota_{X_\alpha} d\tau_k)|_{\mathcal{H}} = (d\lambda_k)|_{\mathcal{H}} \quad \text{and} \quad (\mathcal{L}_{X_\alpha} \omega)|_{\mathcal{H}} = 0.$$

The Jacobi structure is then given by Λ and $E = X_\alpha$.

- We computed the obstruction for a **non-degenerate 2-covariant tensor field** B to define a **Poisson or Jacobi** bracket.
- We **recover all the usual brackets** (from symplectic, lcs, cosymplectic, contact, cocontact...) with a single framework.
- As novel applications, we derived the exact conditions under which almost cosymplectic structures of order p and fat principal bundles induce Poisson or Jacobi brackets.

Further work:

- Investigate if the **dynamical systems** associated to the various geometric structures can also be unified using this formalism.
- Extend these ideas to brackets for **field theories**.

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